

Renewable Energy Systems

Simulation and Layout Analysis of Solar Panels on a Residential Rooftop Using TRNSYS and PVsyst

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Abstract

This study aims to analyze the performance and optimal arrangement of photovoltaic (PV) panels using TRNSYS and PVsyst software. In the first step, the VSUN450-144BMH solar module was simulated in TRNSYS to obtain the I–V and P–V characteristics under various irradiance conditions. In the second stage, the optimal number and arrangement of the panels were determined by manual calculation based on shading constraints and geometric limitations of a rooftop in Tehran. Finally, the PVsyst software was employed to validate and optimize the layout.

1 Introduction

The growing global demand for renewable energy sources has highlighted solar energy as one of the most promising alternatives to fossil fuels. Photovoltaic (PV) systems convert solar radiation directly into electrical energy, offering a clean and sustainable energy solution. However, the real performance of PV systems is affected by several factors such as solar irradiance, cell temperature, shading, and the geometric configuration of the installation.

For precise performance prediction and layout optimization, numerical simulation tools such as TRNSYS and PVsyst are commonly used. In this project, TRNSYS was employed to simulate the electrical behavior of a commercial PV module (VSUN450-144BMH) and to validate the results using catalog data. Subsequently, manual geometric calculations were performed to estimate the maximum number of panels that could be installed on a residential rooftop in Tehran. Finally, PVsyst software was utilized to refine and confirm the optimal arrangement and total system output.

2 System Description

A photovoltaic (PV) panel converts sunlight directly into electricity through the photovoltaic effect. When sunlight hits the surface of the solar cells, made of semiconductor materials such as silicon, the energy from the light photons excites the electrons inside the material. These energized electrons move through an internal electric field, generating a direct current (DC).

The produced DC electricity is then sent to an inverter, which converts it into alternating current (AC) suitable for household or grid use. The amount of generated electricity depends on solar irradiance, cell temperature, and the orientation of the panel. In summary, a solar panel simply captures sunlight and turns it into usable electrical power without any moving parts or fuel consumption.

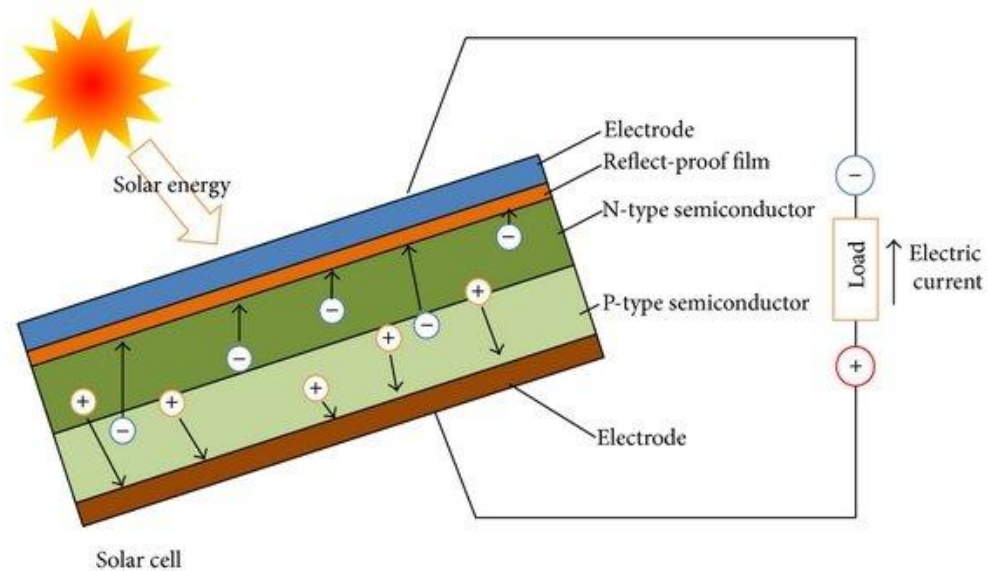


Figure 1. Schematic figure of how PV panel works¹

The PV system was designed for a residential rooftop located in Tehran, Iran. The panels were oriented southward with a tilt angle of 35° , which is slightly higher than the latitude of the site (32°) to prevent snow accumulation on the surface. According to PVsyst analysis, the optimal tilt angle for maximum annual yield was between 32° and 36° , confirming the chosen configuration.

¹ www.researchgate.net

The rooftop configuration allowed a combination of vertical and horizontal arrangements to maximize surface coverage while maintaining appropriate shading distances. The system consisted of 21 VSUN450-144BMH panels.

3 Modeling

3.1 Modeling of the PV Module in TRNSYS

module was created based on the parameters provided in the manufacturer's catalog. All technical data, including open-circuit voltage, short-circuit current, nominal operating conditions, and temperature coefficients, were entered into the PV component. By applying voltage as an input variable to the model, the I-V and P-V characteristics of the panel were obtained under five different irradiance levels (1000, 800, 600, 400, and 200 W/m²).

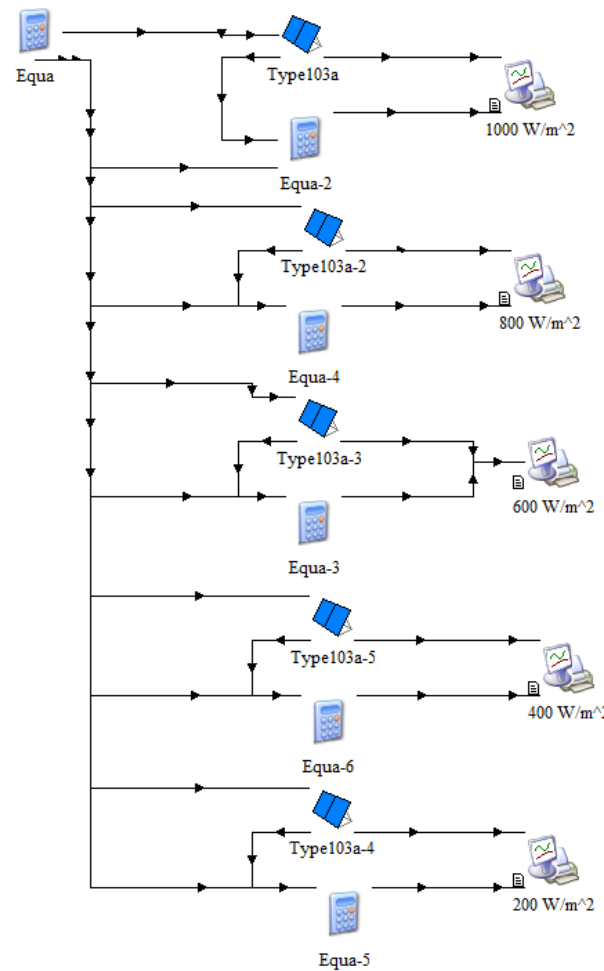


Figure 2. TRNSYS schematic of the PV system

The corresponding I–V and P–V curves obtained from the simulation are shown below:

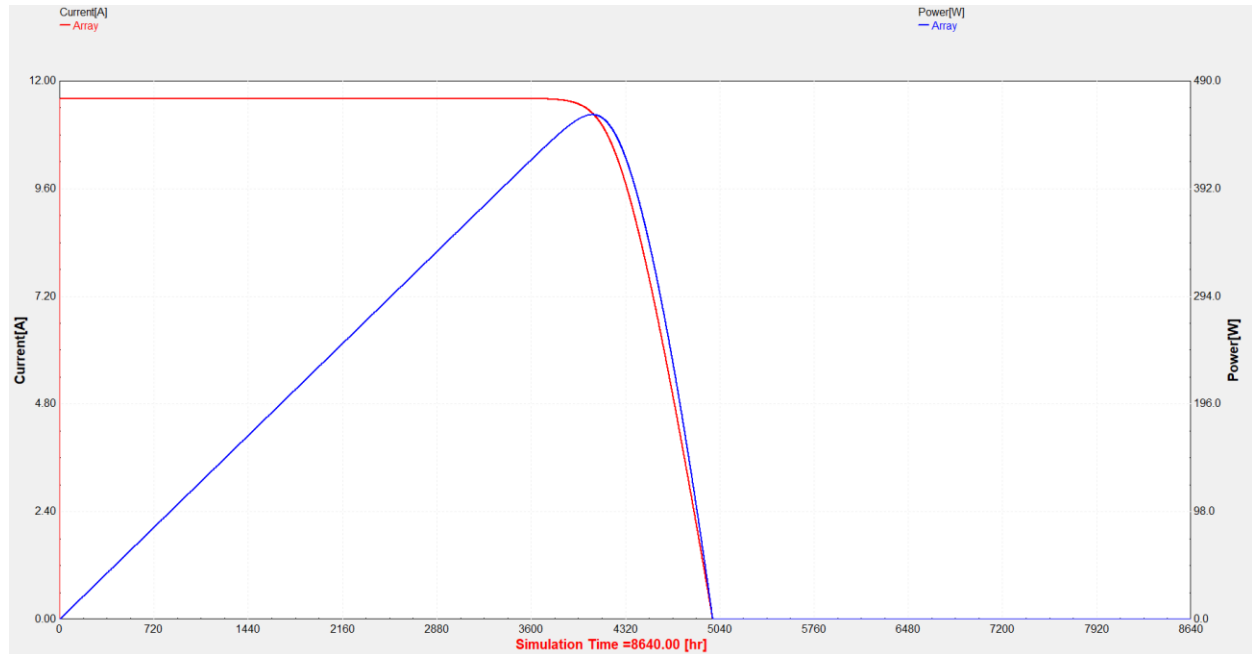


Figure 3. I–V and P–V characteristics of module under irradiance 1000 W/m^2

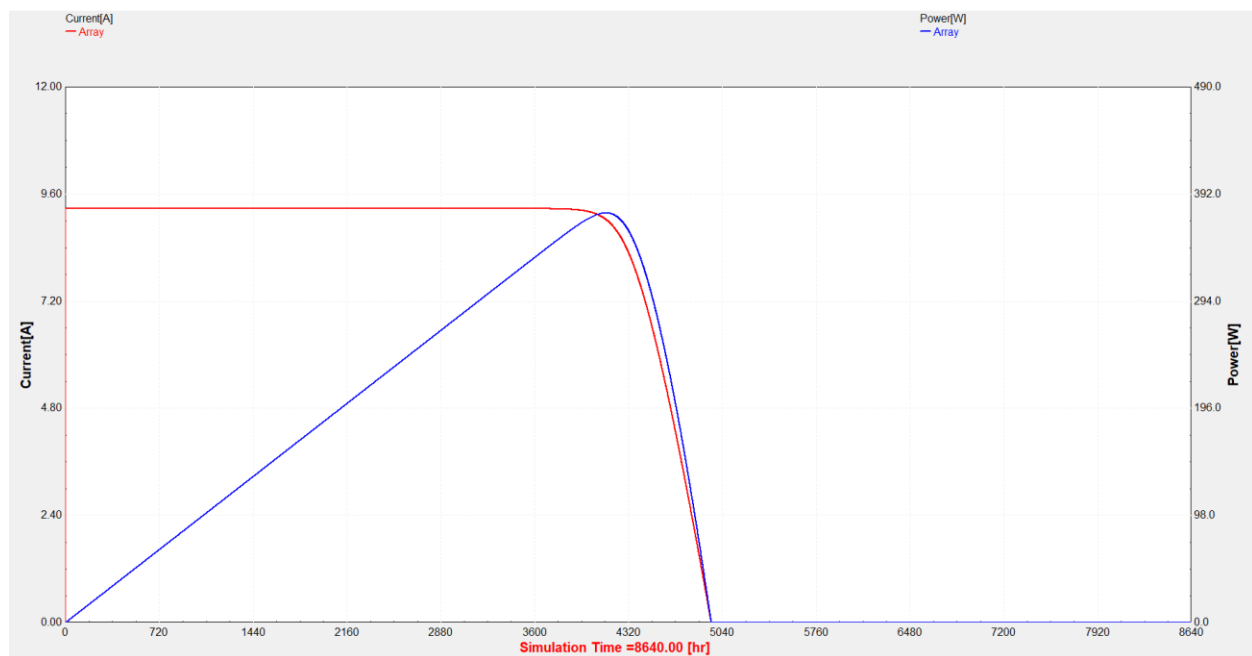


Figure 4. I–V and P–V characteristics of module under irradiance 800 W/m^2

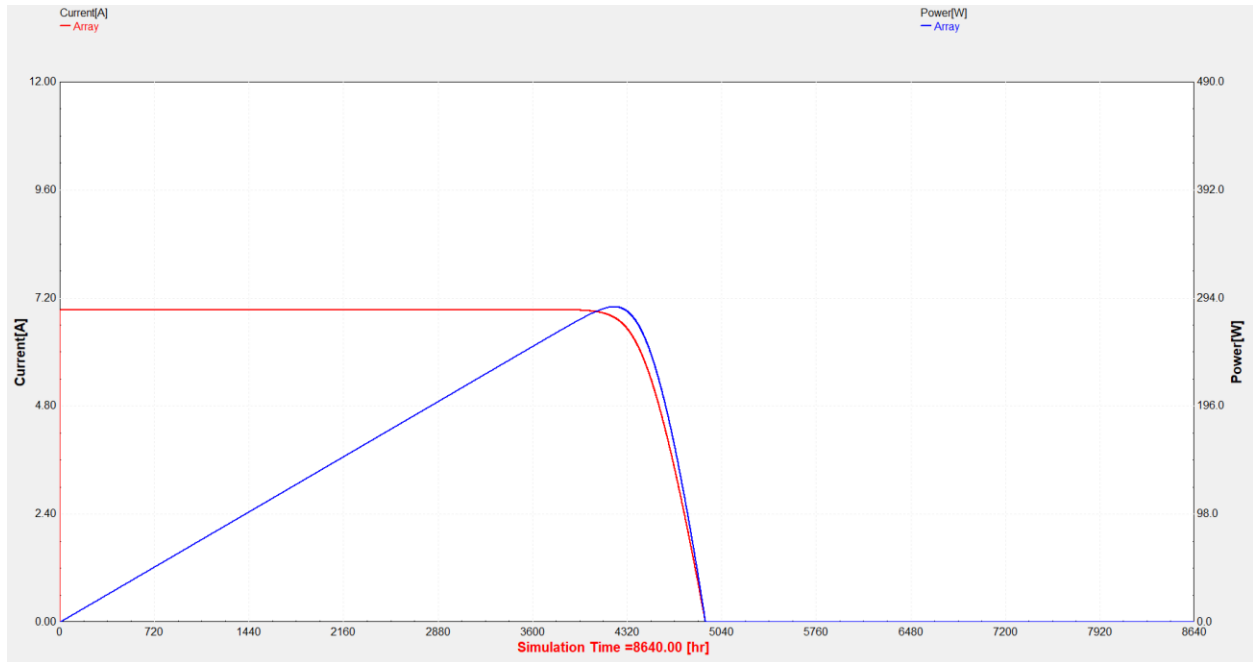


Figure 5. I–V and P–V characteristics of module under irradiance 600 W/m^2

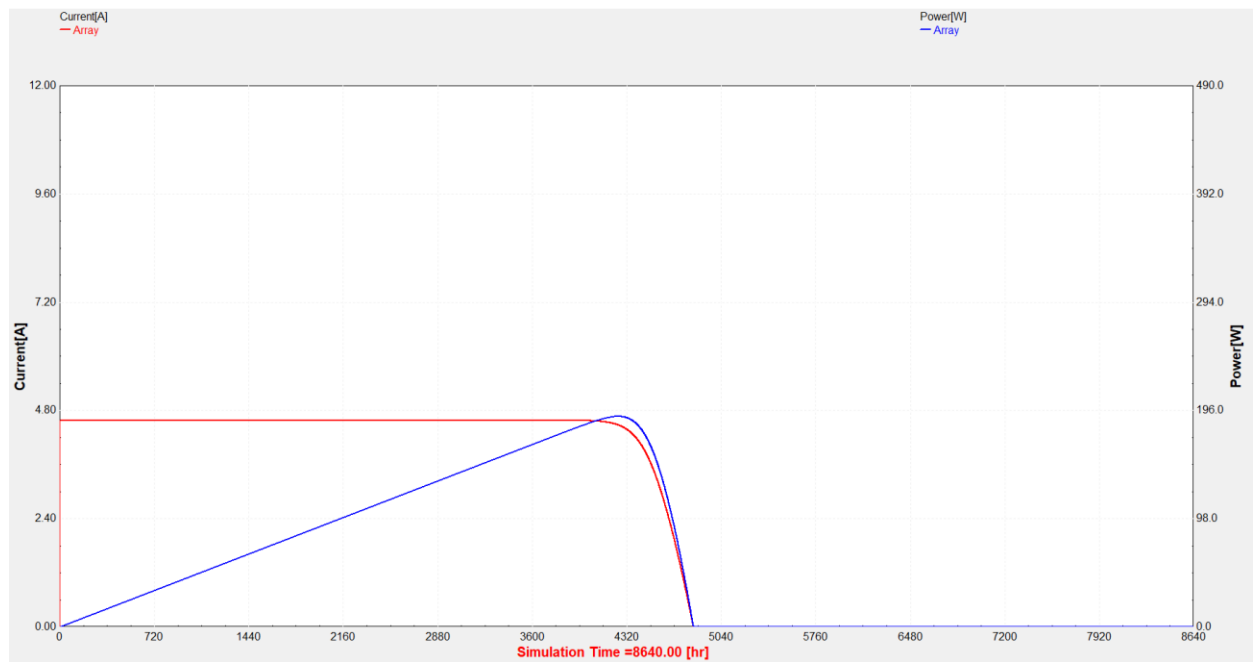


Figure 6. I–V and P–V characteristics of module under irradiance 400 W/m^2

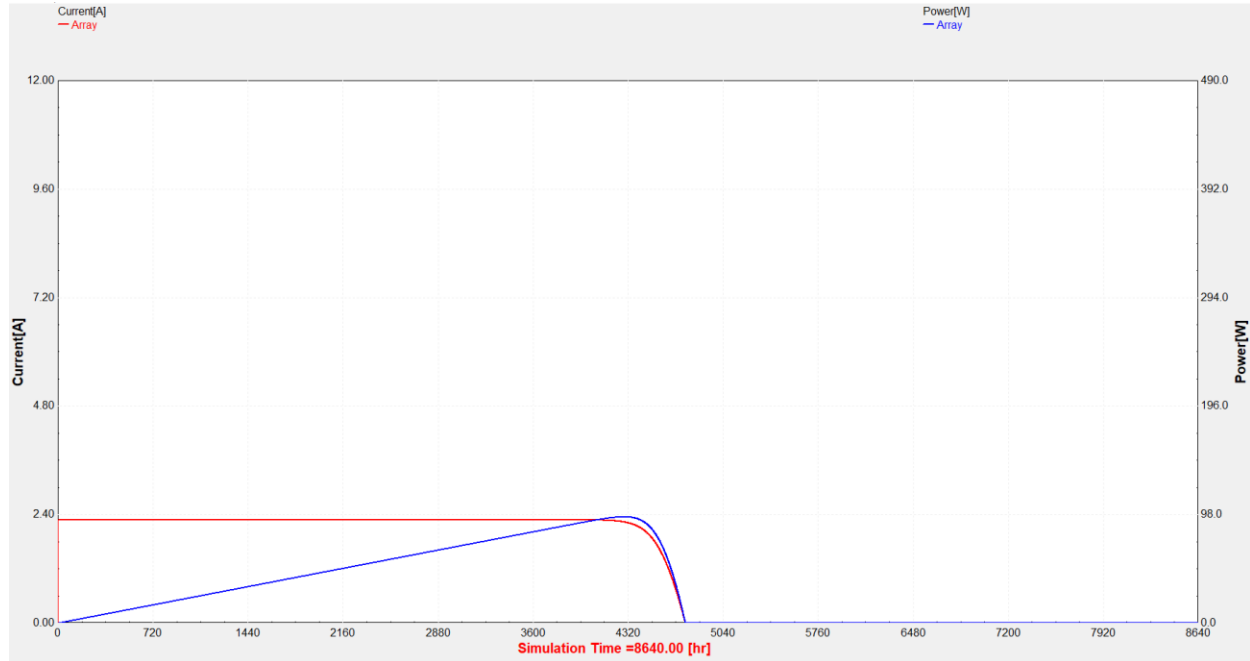


Figure 7. I–V and P–V characteristics of module under irradiance 200 W/m²

The accuracy of the TRNSYS model was validated by comparing the simulated I–V curves with the manufacturer’s catalog data. The overall trend of the simulated curves matched the reference data, with the maximum deviation of 6.5%, occurring near the maximum power point at 1000 W/m².

The sources of this deviation can be attributed to:

- Simplified modeling assumptions.
- Uncertainties in temperature coefficient values.
- Neglecting minor resistive losses and contact temperature effects.
- Measurement and rounding errors in catalog data.

Despite these discrepancies, the correlation between simulated and reference curves was satisfactory, confirming that the TRNSYS model accurately represents the panel’s electrical behavior.

3.2 Radiation, Temperature, and Power Simulation for Tehran

To evaluate the annual performance of the PV system, the climatic data for Tehran were imported from the Meteonorm weather database available in TRNSYS. These data include hourly values of global solar radiation, ambient temperature, and wind speed.

By connecting these weather data to the PV module component, the cell temperature, incident radiation on the tilted surface, and instantaneous output power were calculated for the entire year. The simulation also provided the monthly average power production, which represents the variation of generated energy throughout different seasons.

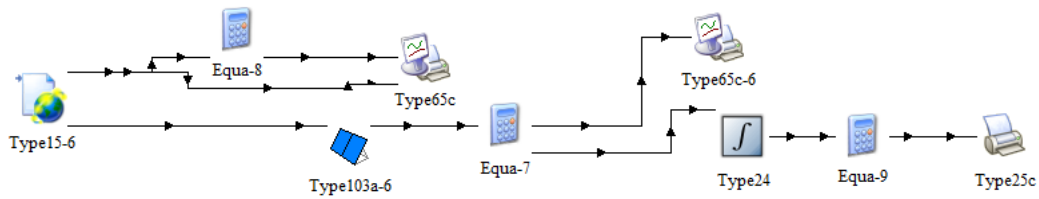


Figure 8. TRNSYS schematic of weather and the PV system

The results clearly show higher power production during summer months due to increased irradiance, and lower output during winter when solar radiation intensity are reduced.

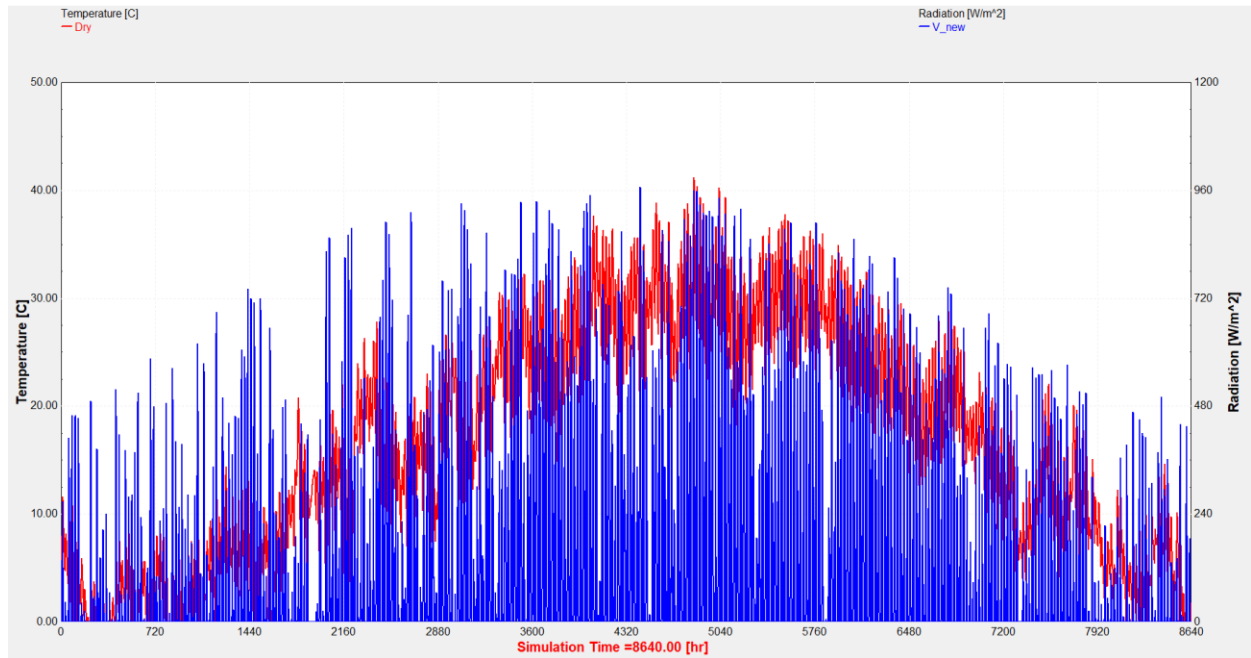


Figure 9. Hourly variation of irradiance, module temperature, and PV power output for Tehran

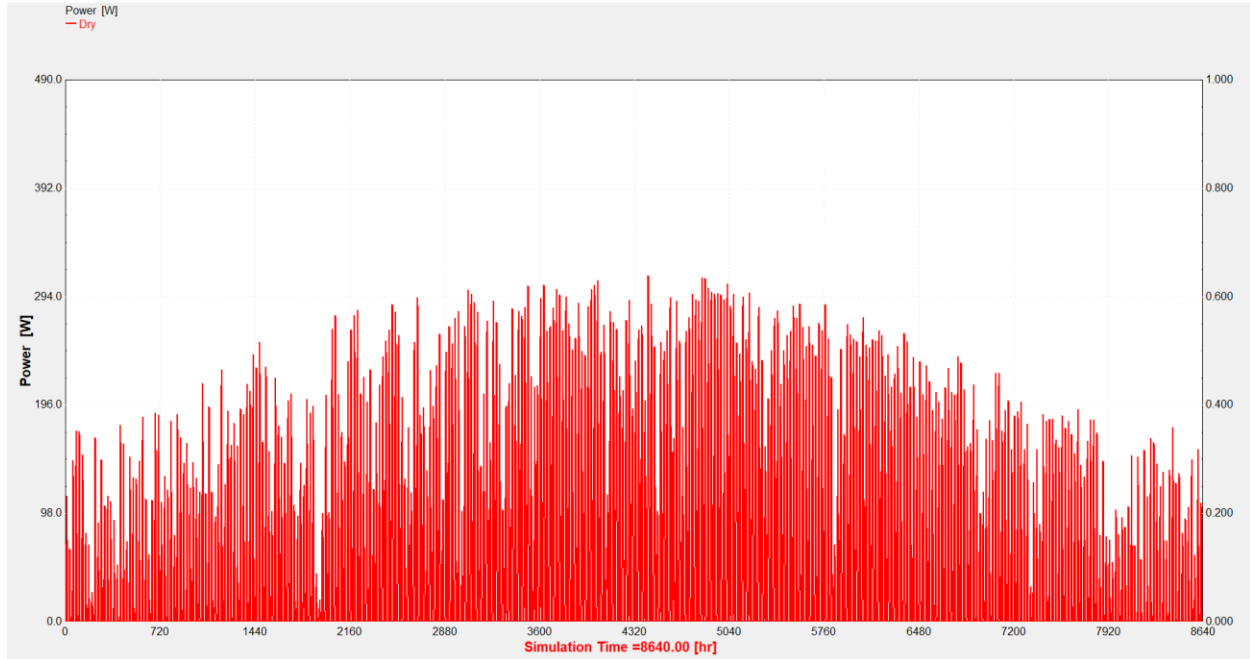


Figure 10. PV Panel power in Tehran climate throughout the year

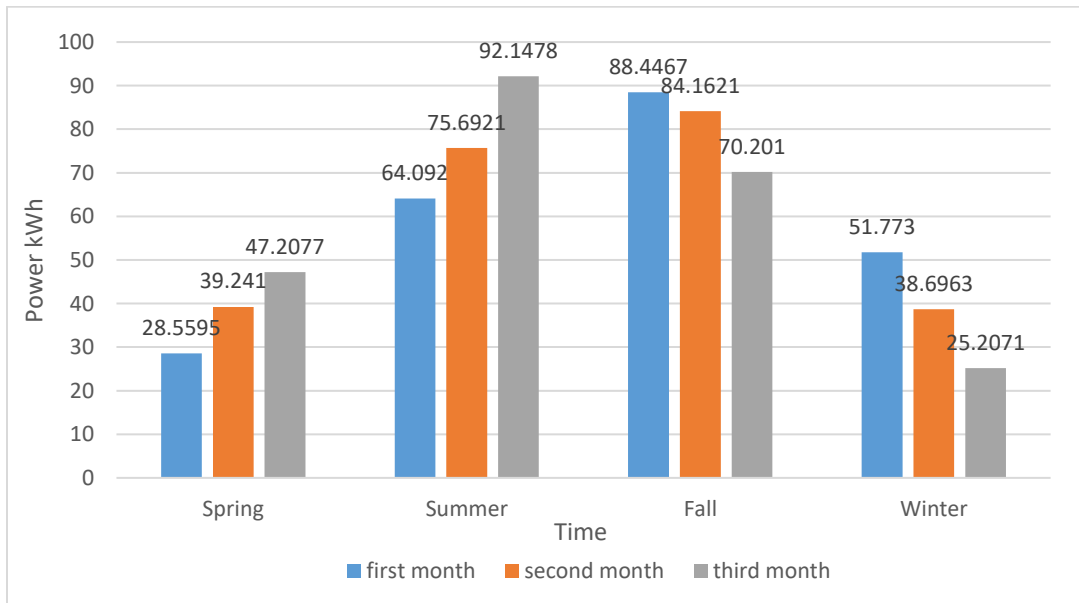


Figure 11. Monthly average power production of the PV system

3.3 Estimate the number of panels

The minimum spacing between rows was calculated using Eq. (1), based on the solar elevation angle (α) at 3 p.m. on the winter solstice (December 21). To calculate the angles of the sun in the worst case, we used the website [andrewmarsh](http://andrewmarsh.com).

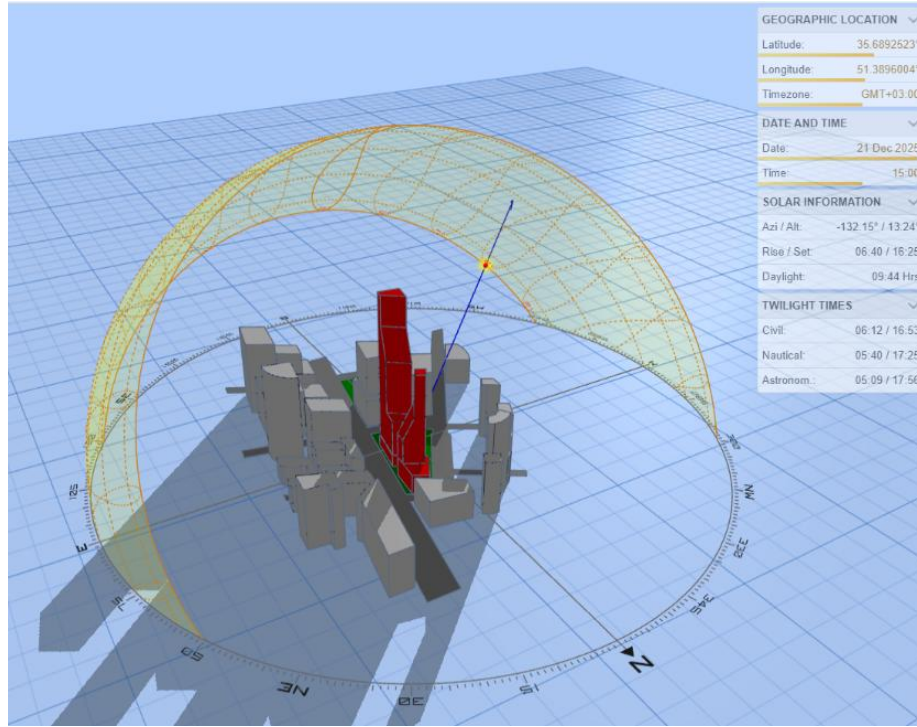


Figure 12. Geographical characteristics of Tehran²

$$\frac{L \times \sin \beta \times \cos \gamma}{\tan \alpha} = D_{min} \quad (1)$$

Table 1. Parameters value

parameter	Value	Unit
L (the panel width)	1.048 , 2.108	m
β (the panel tilt angle)	35	degree
γ (the azimuth)	42.15	degree
α (the altitude angle)	13.24	degree
D (The distance between two panels)	wanted	m

According to this equation, D is equal to 3.81 in vertical mode and equal to 1.894 m in horizontal mode. The roof that is supposed to be placed on those panels is as follows. In the middle, there is a truss measuring 2 m × 2 m and height 1 m. It can be simulated like a horizontal panel to calculate the distance of the panel from behind it.

² andrewmarsh.com

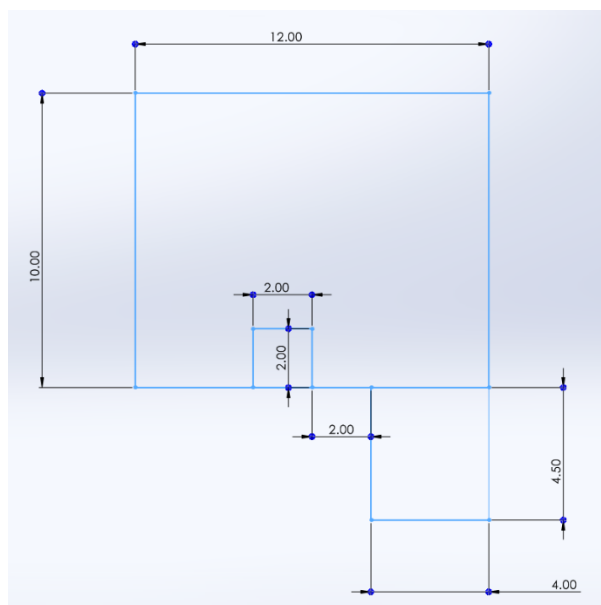


Figure 13. The shape of the roof surface

Both horizontal and vertical orientations of the PV panels were analyzed in PVsyst to evaluate the impact of geometry on the available rooftop area and shading performance. The results showed that the combined configuration, which included a mix of horizontally and vertically mounted panels, allowed a larger number of modules to be installed within the same surface area compared to using a single orientation. Also, based on the relevant standards (MCS MIS 3002), the distance of the panels from the sides of the roof is considered equal to 30 cm.

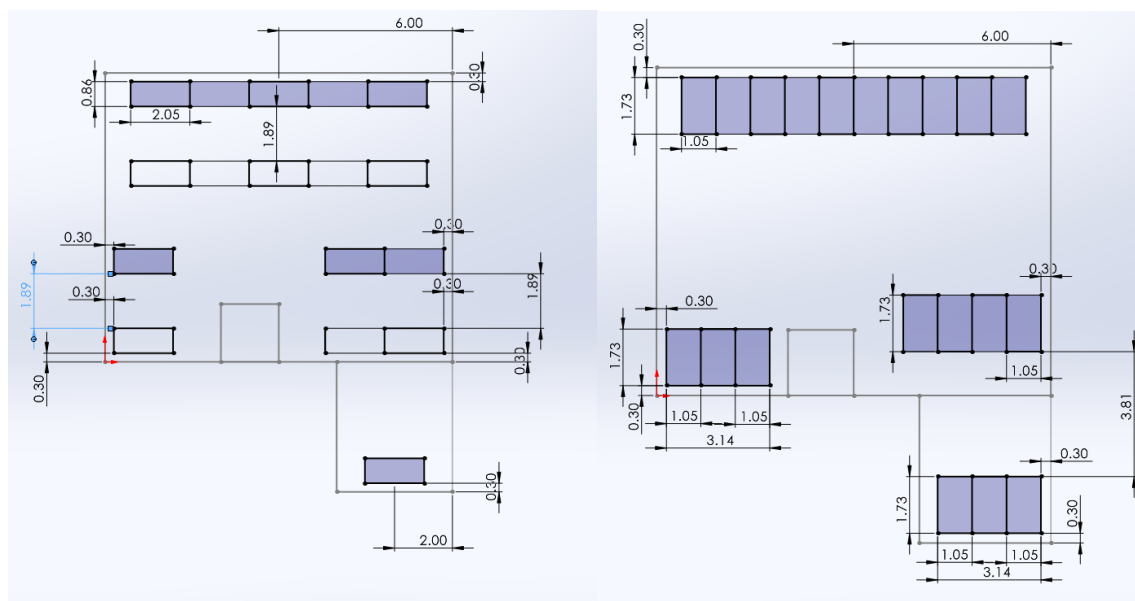


Figure 14. horizontal and vertical PV panels on the rooftop

However, to minimize mutual shading effects, special attention was given to the arrangement of the panels. In particular, no horizontal panel was placed directly behind a vertically mounted one on the right-hand side, as this position would cause shading during afternoon hours when the solar elevation angle is low.

This optimized hybrid arrangement achieved an effective balance between maximizing the number of installed panels and minimizing shadow-induced energy losses. One of the main challenges was choosing the right number of panels in a pack so that they could be placed at the bottom of the roof.

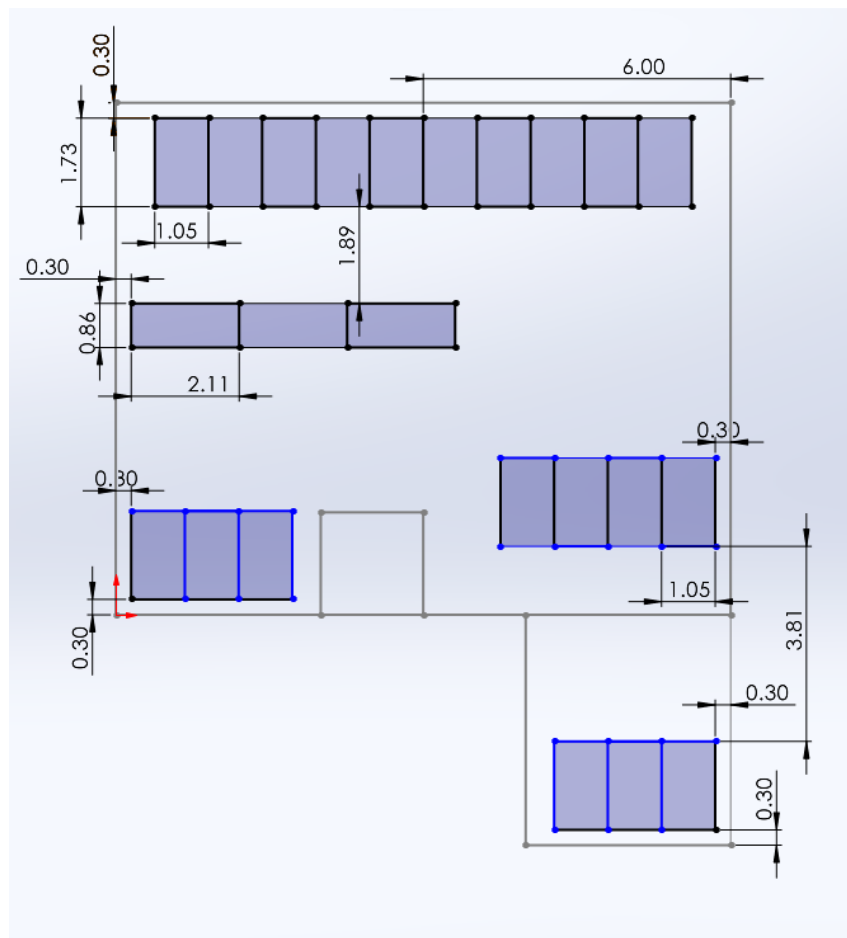


Figure 15. The position of the PV panels on the roof

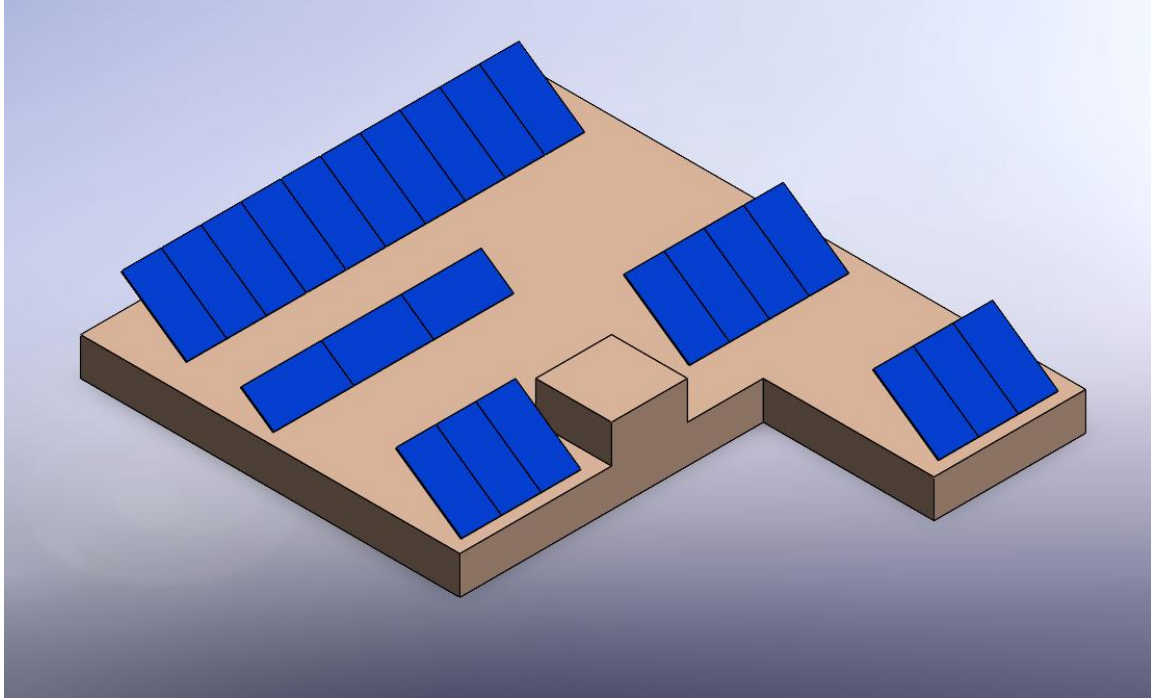


Figure 16. 3D plan of roof and panels

3.4 PVsyst Modeling and Layout Analysis

In the PVsyst software, the geographical location of the installation site was first defined using the latitude and longitude coordinates of Tehran. After locating the site, the tilt angle of the panels (35°) and the azimuth orientation toward the south (0°) were specified.

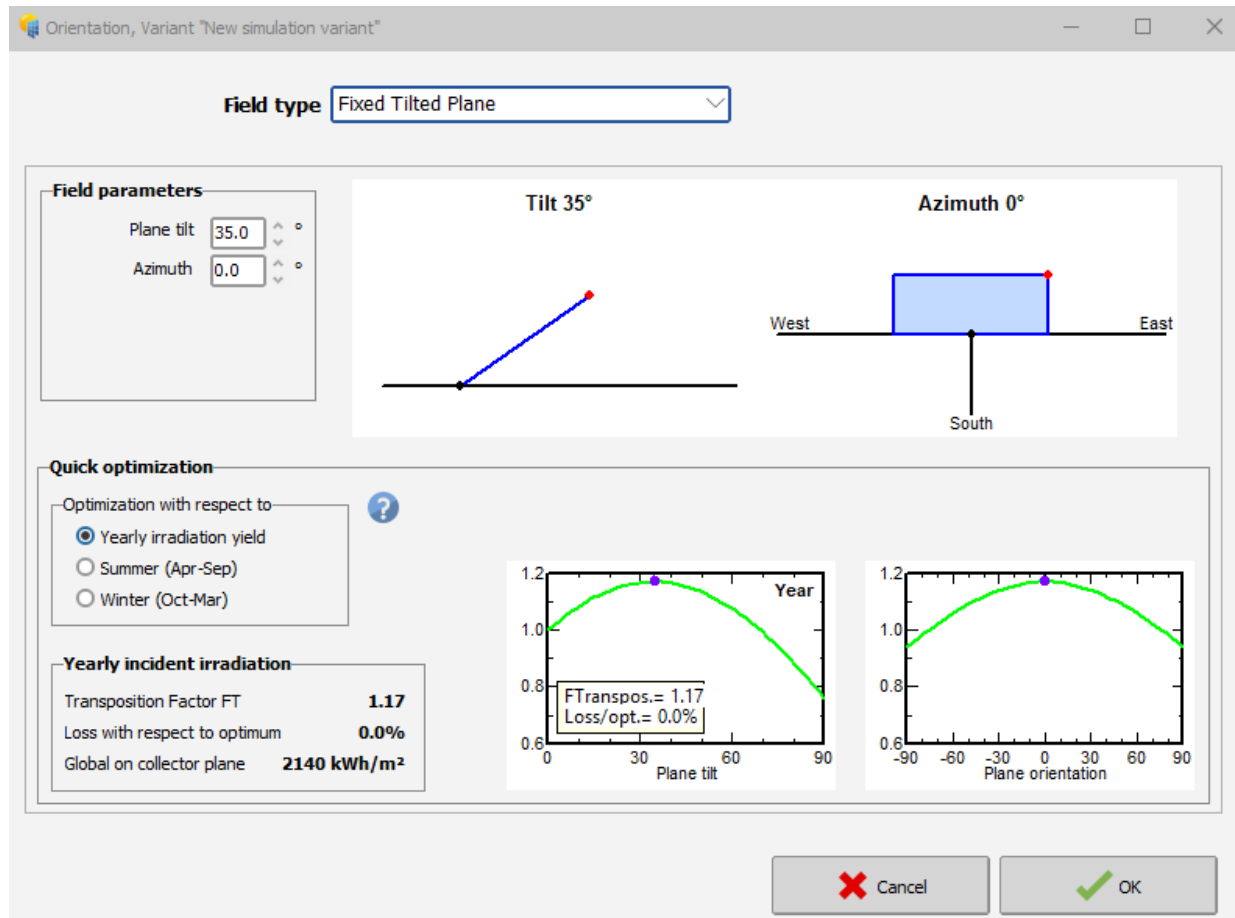


Figure 17. orientation in PVsyst

At the initial stage of the PVsyst simulation, a 3D layout of the system was created in PVsyst, including the panels and the underlying building structure. This step was necessary to accurately determine the total number of panels that could be installed within the available surface area and to define the number of possible electrical strings. The selected PV module, VSUN450-144BMH, was available in the PVsyst database and was therefore chosen directly from the software's library. The 3D shading scene was analyzed to ensure that the beam linear loss due to mutual shading between rows was less than 1%, thereby maintaining high system efficiency.

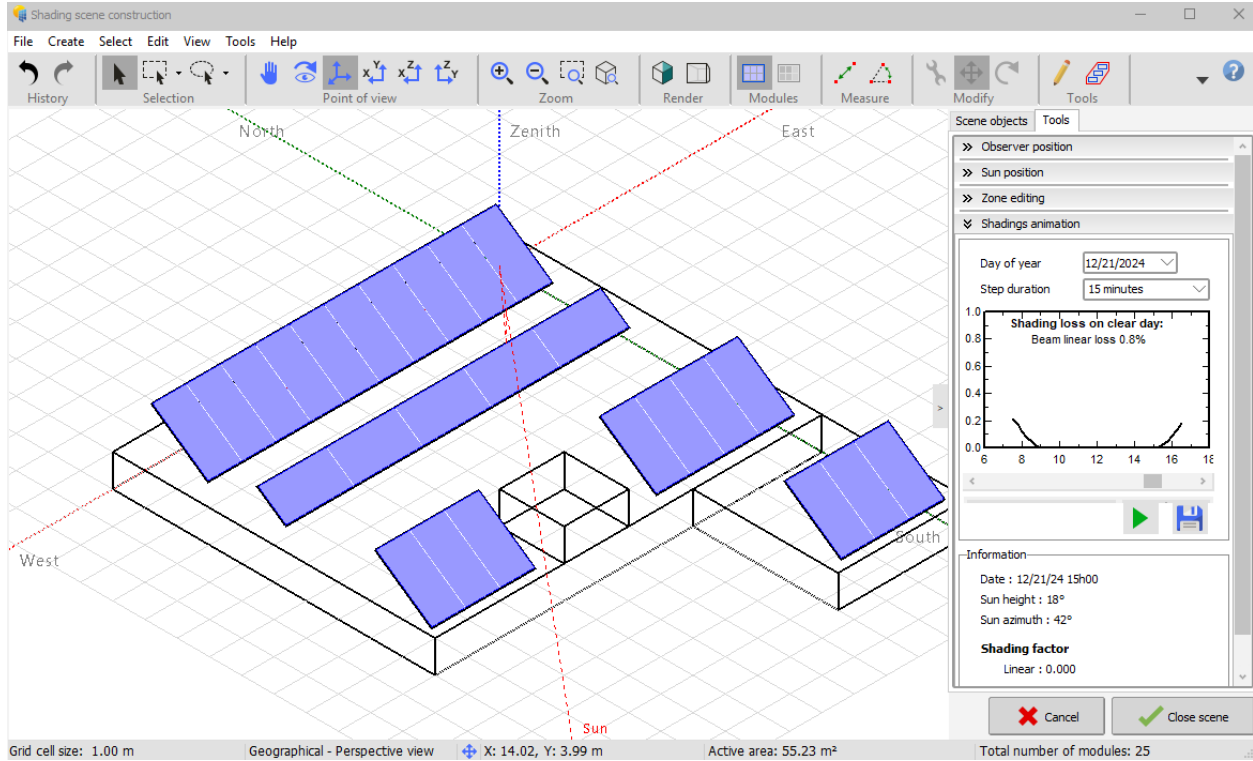


Figure 18. construction in PVsyst

After completing the 3D layout, the total number of panels was found to be 25 modules, which corresponded to five possible strings based on the module interconnection configuration. Subsequently, a search was performed in the Huawei inverter catalog to identify compatible models capable of operating within the required voltage and power range. Among the available options, only the Huawei 2 kW TL inverter (80–600 V) met the system requirements and provided appropriate voltage compatibility with the panel’s electrical characteristics.

Therefore, this inverter was selected for the final configuration, ensuring reliable operation and optimal energy conversion efficiency. The operating voltage range, obtained from the manufacturer’s catalog, extended down to approximately 30 V, which was considered during the inverter selection process.

Grid system definition, Variant VCO: "New simulation variant"

Sub-array

Sub-array name and Orientation
Name:
Orient.: **Fixed Tilted Plane**
Tilt: **35°**
Azimuth: **0°**

Pre-sizing Help
☒ No sizing
Planned power: kWp
... or available area(modules): m²

Select the PV module
Available Now:
Filter:
450 Wp 35V Si-mono VSUN-450-144-MH Since 2020 Datasheets 2020
☐ Use optimizer
Sizing voltages : Vmpp (60°C) **35.6 V**
Voc (-10°C) **55.4 V**

Select the inverter
Available Now:
Output voltage 230 V Mono 50Hz
2.0 kW 80 - 600 V TL 50/60 Hz SUN2000-2KTL-L1 Since 2020
Nb. of inverters:
☐ Use multi-MPPT feature
Operating voltage: **80-600 V** Global Inverter's power: **10.0 kWac**
Input maximum voltage: **600 V** **inverter with 2 MPPT**
Power sharing within this inverter

Design the array
Number of modules and strings
Mod. in series: ☐ between 3 and 10
Nb. strings:
Overload loss: **0.1 %**
Pnom ratio: **1.13**
Nb. modules: **25** Area: **55 m²**

Operating conditions
Vmpp (60°C) 178 V
Vmpp (20°C) 211 V
Voc (-10°C) 277 V
Plane irradiance: **1000 W/m²**
Impp (STC) 54.3 A
Isc (STC) 57.5 A
Isc (at STC) 57.5 A
Max. operating power (at 1116 W/m² and 50°C) **11.4 kW**
Array nom. Power (STC) **11.3 kWp**

List of subarrays

Name	#Mod #Inv.	#String #MPPT
PV Array		
VSUN - VSUN-450-144-MH	5	5
Huawei Technologies - SUN2000...	5	1

Global system summary
Nb. of modules: 25
Module area: 55 m²
Nb. of inverters: 5
Nominal PV Power: 11.3 kWp
Nominal AC Power: 10.0 kWAC
Pnom ratio: 1.125

Figure 19. system in PVsyst

Finally, the simulation of the configured system produced the expected electrical output, from which the annual energy yield and performance ratio were obtained. The report file of this software is attached. The annual energy output was estimated at 19,255 kWh/year, and the system achieved a performance ratio of approximately 81.4%, which is acceptable for fixed-tilt rooftop installations.

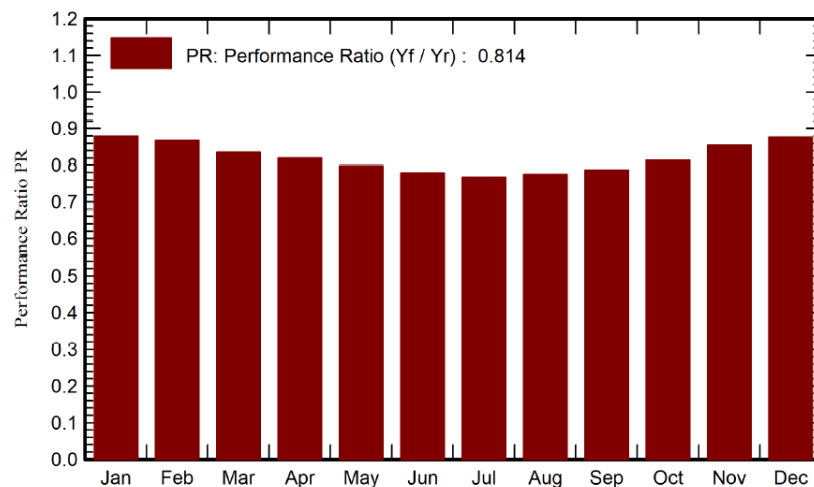


Figure 20. Performance Ratio

4 Results and Discussion

The simulated I–V and P–V curves showed the expected nonlinear behavior, where an increase in solar irradiance caused a proportional rise in current and power, while the open-circuit voltage increased slightly.

Annual simulation results indicated that:

- Maximum energy generation occurred during the summer months when irradiance was highest.
- Minimum production was observed in December and January due to lower solar elevation angles.

The combination of vertical and horizontal panel orientation optimized both shading reduction and surface coverage, providing a balanced compromise between mechanical stability and energy performance.

5 Conclusion

This study demonstrated an integrated approach for analyzing and optimizing rooftop PV systems using TRNSYS and PVsyst software.

The TRNSYS simulation successfully reproduced the electrical characteristics of the VSUN450-144BMH module under different irradiance levels, with a maximum deviation of 6.5% from catalog data.

The PVsyst analysis confirmed that a total of 21 panels, each rated at 450 W, could be installed on the selected rooftop, achieving an overall nominal capacity of about 19,255 kWh/year, which is acceptable for fixed-tilt rooftop installations.

The slightly increased tilt angle (35° instead of 32°) not only improved snow shedding but also remained within the optimal range ($32\text{--}36^\circ$) predicted by PVsyst.

Minor discrepancies between simulated and catalog data were mainly due to simplifications in the TRNSYS model, nonuniform temperature distribution on the panel surface, and minor parameter uncertainties.

Overall, the project confirms that using TRNSYS for electrical modeling and PVsyst for spatial optimization provides a comprehensive and accurate methodology for the design and analysis of small-scale rooftop PV systems in urban environments like Tehran.

6 References

MCS MIS 3002 — Installation of Solar Photovoltaic Systems.

TRNSYS 18 User Manual, Thermal Energy System Specialists, 2020.

PVsyst 7.3 Help Documentation, PVsyst SA, 2023.

VSUN Energy Co., Ltd. — VSUN450-144BMH Product Datasheet.